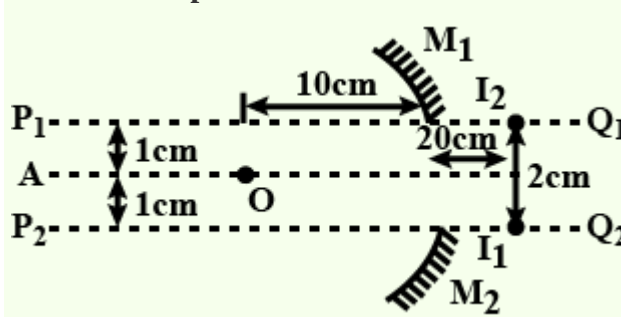
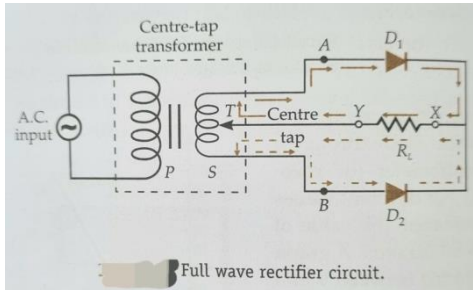
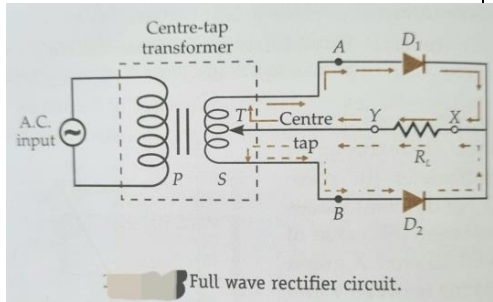
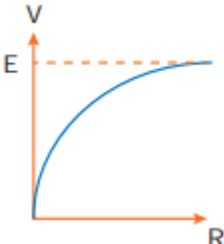
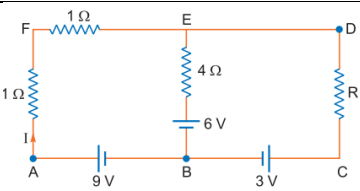
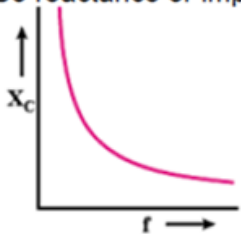
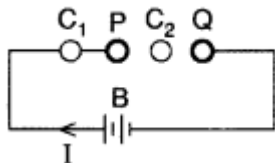


Q NO		MARKS
	SECTION A	
1	(c) The dipole will experience a force towards left.	1
2	(a) zero	1
3	(a)	1
4	(c)	1
5	(d) Using expression For short dipole $B_{equatorial} = \frac{\mu_o M}{4\pi r^3}$ 0.48 G along N-S direction	1
6	(b) Induced current is anticlockwise in the loop.	1
7	(d) All electromagnetic waves travel with the speed of light in space.	1
8	(a)	1
9	(d) $8 \lambda_1 D/d = 9 \lambda_2 D/d \Rightarrow 8 \lambda_1 = 9 \lambda_2$	1
10	(d) No photoelectric emission happens when the frequency of incoming radiation falls below a specific threshold	1
11	(d)	1
12	(a) 2 mm Solution The correct option is A 2 mm  Let P1Q1= principal axis for the mirror M1,	1

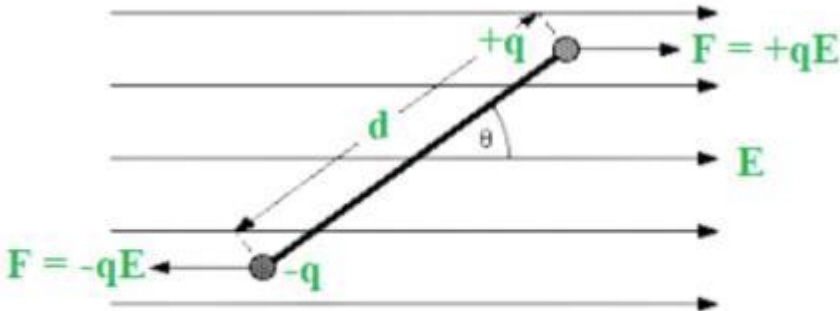
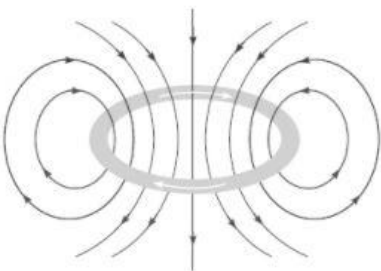
	P2Q2= principal axis for the mirror M2 (I1= Image formed by M1, I2= Image formed by M2) Height of object and image are measured with respect to principal axis of respective mirror. Applying sign convention, we have, $f = -20 \text{ cm}, u = -10 \text{ cm}$ and by Mirror formula $\frac{1}{-10} + \frac{1}{v} = \frac{1}{-20} \Rightarrow v = 20 \text{ cm}$ Magnification of image is $m = -\frac{v}{u} = -\frac{20}{-10} = 2 = \frac{h_i}{h_o} \Rightarrow h_i = 2 \text{ cm}$ (+ve magnification means image and object are on the same side with respect to principal axis.) $h_{I1} = 2 \text{ cm}$ below the principal axis P1Q1 $h_{I2} = 2 \text{ cm}$ below the principal axis P2Q2 $\Rightarrow I1I2 = 2 \text{ cm}$	
13	(a)	1
14	(b)	1
15	(b)	1
16	(d)	1
	SECTION B	
17	X- half wave rectifier. Y-full wave rectifier.  	1
18	Total energy. $E = -13.6 \text{ eV}$ $\text{K.E.} = -E = 13.6 \text{ eV}$	1 1

	P.E.= -2 K.E. = -27.2 eV	
19	Critical angle(C) = 45° . Speed in medium (v)=? $\mu = \frac{3 \times 10^8}{v} = \frac{1}{\sin 45^\circ}$ $v = \frac{3 \times 10^8}{\sqrt{2}} = 2.12 \times 10^8 \text{ m/s}$ Yes, correct reason Or If $\Delta = \frac{\lambda}{3}$ And we know $\phi = \frac{2\pi}{\lambda} \Delta = \frac{2\pi}{\lambda} \frac{\lambda}{3} = \frac{2\pi}{3}$ And $I = I_0 \cos^2 \frac{\phi}{2} = I_0 \cos^2 \frac{2\pi}{6} = I_0$	1 0.5+0.5 1 1
20	$S = IG / (I - I) = 0.12 \Omega$ Net resistance	1 1
21	$V = E - ir$ $V = IR = \frac{E}{(R+r)} R = \frac{E}{(1+\frac{r}{R})}$ 	1 1
	SECTION C	
22.	Applying Kirchhoff's loop rule for loop ABEFA, $-9 + 6 + 4 \times 0 + 2I = 0$ or, $I = 1.5 \text{ A}$ For loop BCDEB, $3 + IR + 4 \times 0 - 6 = 0$ or, $R = 2 \Omega$ To find Potential difference between A and D, using KVL through open loop ABCD, $9 - 3 - IR = V_A - V_D = 3V$ 	1 1 1

23.	Correct Explanation Or $C = \epsilon_0 A/d$ (i) when battery is not disconnected, then $V = \text{constant}$, $d' = 3d$ & $K = 10$ $C' = K \epsilon_0 A/d' = 10 \epsilon_0 A/3d = 10 C/3$ (ii) Charge $Q' = C'V = 10/3 CV = 10Q/3$ (iii) Energy density, $u' = 10 u/9$	1 1 1
24.	(i) Device X is a capacitor. As the current is leading voltage by $\pi/2$ (ii) Curve A represent power, Curve B represents voltage and curve C represents current. As $I = I_m \sin \omega t$ $V = V_m \sin \omega t$ As in the case of capacitor, $i = i_m \sin \left(\omega t + \frac{\pi}{2} \right)$ Average power, $P = V_i = V_m i_m \cos \phi / 2$ (iii) as, Capacitive reactance, $X_C = \frac{1}{\omega C}$ So reactance or impedance decreases with increase in frequency.  OR Angular frequency, $\omega = \frac{1}{\sqrt{LC}}$ (i) $\omega = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = \frac{1}{\sqrt{400 \times 10^{-6}}}$ $= \frac{1}{20 \times 10^{-3}} = \frac{1000}{20} = 50 \text{ radians/sec.}$ (ii) $I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{240}{40} \text{ A} = 6 \text{ A}$ (iii) V_{rms} across capacitor $= 6 \times \frac{1}{\omega C}$ $= 6 \times \frac{1}{50 \times 80 \times 10^{-6}} \text{ V}$ $= \frac{6 \times 10^6}{4 \times 10^3} \text{ V} = \frac{6000}{4} \text{ V} = 1500 \text{ V}$	1 1 1 1 1
25.	a. Microwaves are used in radar systems that are used in aircraft navigation. These rays can be produced by making use of special vacuum tubes, such as Klystrons, magnetrons and Gunn diodes etc.	1

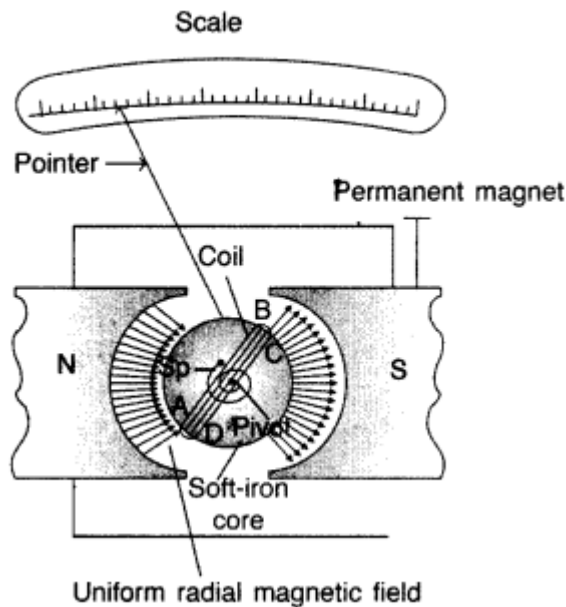
	b. Infrared waves are used in the treatment of muscular strain. These rays can be produced by hot bodies and molecules. c. X rays are used as a very significant diagnostic tool in medicine. These rays are produced when high energy electrons hit any metal of a greater atomic number.	1 1
26.	(a) The momentary deflection is due to the transient current flowing through the circuit when the capacitor is getting charged. The deflection would be zero when the capacitor gets fully charged. (b) Maxwell's displacement current : According to Ampere's circuital law, the magnetic field B is related to steady current I as  Maxwell showed that this relation is logically in-consistent. He accounted this inconsistency as follows : Ampere's circuital law for loop C_1 gives $\oint_{C_1} \vec{B} \cdot d\vec{l} = \mu_0 I \quad \dots(ii)$ Loop C_2 lies in the region between the plates $\therefore \oint_{C_2} \vec{B} \cdot d\vec{l} = 0 \quad \dots(iii)$ It is expected that, $\oint_{C_1} \vec{B} \cdot d\vec{l} = \oint_{C_2} \vec{B} \cdot d\vec{l} \quad \dots(iv)$ which is logically inconsistent. So, Maxwell gave idea of displacement current. Thus displacement current is that current which comes into play in the region in which the electric field and hence the electric flux is changing	1 1

	with time. $I_D = \epsilon_0 \frac{d\phi E}{dt}$ where I_D is displacement current and ϕE is electric flux. $= \mu_0 \left(I + \epsilon_0 \frac{d\phi E}{dt} \right)$ It is now called Ampere-Maxwell law. This is the generalization of Ampere's circuital law	1
27.	(i) Distance between two dark lines, on either side of central maxima = $2 \frac{\lambda D}{a}$ $= \frac{2 \times 6000 \times 10^{-10} \times 1.5}{1 \times 10^{-4}} = 18000 \times 10^{-6}$ $= 18 \times 10^{-3} \text{m} = \mathbf{18 \text{ mm}}$ (ii) Angular spread of the first diffraction minimum (on either side) $= \theta = \frac{\lambda}{a} = \frac{6 \times 10^{-7}}{1 \times 10^{-4}} = \mathbf{6 \times 10^{-3} \text{ radians}}$	1 1 1
28.	According to Einstein's photoelectric equation, $V_0 = \frac{h}{e} \nu - \frac{\phi_0}{e}$ In the given graph : Stopping potential, $V_0 = 1.23 \text{ V}$ Change in frequency, $\Delta \nu = (8 \times 10^{14} - 5 \times 10^{14})$ $= 3 \times 10^{14} \text{ Hz}$ $\therefore$ Slope of the line = $\frac{h}{e}$ $\therefore \frac{h}{e} = \frac{V_0}{\Delta \nu} = \frac{1.23}{3 \times 10^{14}}, \therefore e = 1.6 \times 10^{-19} \text{ C}$ $\therefore h = \frac{1.23 \times 1.6 \times 10^{-19}}{3 \times 10^{14}} \text{ JS} = 6.6 \times 10^{-34} \text{ JS}$	1 1 1
	SECTION D	
29	1.(a) $\tau = \mathbf{P \times E}$; 2.(c) $2 \times 10^{-3} \text{ Nm}$ (As maximum torque is given by, $\tau_{\text{max}} = PE$, $\sin \theta \square \square \text{max. value} = 1$) 3.(a) 0°; Torque = $PE \sin \theta$ and $\sin \theta$ has minimum value equal to 0, for $\theta = 0^\circ$	1 1 1 1

	4. (c) $F = 0, \tau \neq 0$; As from figure, it is clear that under the given condition two equal, opposite and parallel forces begin to act on the dipole forming a couple and due to this couple a torque is generated on the dipole. But at the same time we notice that the two equal and opposite forces also act on the dipole making thereby net force equal to zero.  Electric Dipole in External Field OR 4. (a) $pE\sin\theta, -pE\cos\theta$; Torque = $pE\sin\theta$ and potential energy of the dipole = $-\mathbf{P} \cdot \mathbf{E} = -pE\cos\theta$	1
30.	1. (b) 2.(a) 3. (a) 4. (a) OR 4. (b)	1 1 1 1 1
	SECTION E	
31.	(a)  (b) Moving coil galvanometer. It is a device used for the detection and measurement of small electric current. Principle. The working is based on the fact that a current carrying coil	1

suspended in a magnetic field experiences a torque.

Construction. It consists of a coil having a large number of turns of insulated copper wire wound on a metallic frame. The coil is suspended by means of a phosphor-bronze strip and is surrounded by a magnet N S. A hair spring is attached to lower end of the coil. The other end of the spring is attached to the scale through a pointer.



Working. When current is passed, say along ABCD, the couple acts on it. Since the plane remains always parallel to the magnetic field in all positions of the coil (radial field), the force on the vertical arms always remains perpendicular to the plane of the coil.

Let, I be the current flowing through coil,

B be magnetic field supposed to be uniform and always parallel to the coil,

A be area of the coil

Deflecting torque acting on the coil is,

$$\tau = nI BA \sin 90^\circ$$

Due to deflecting torque, the coil rotates and suspension wire gets twisted.

A restoring torque is set up in the suspension fibre. If ϕ is angle through which the coil rotates and k is the restoring torque per unit angular twist, then restoring torque, $\tau = k\phi$

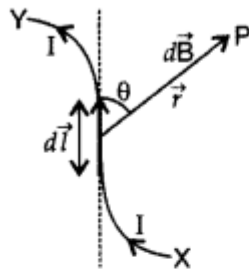
In equilibrium,

	Deflecting torque = Restoring torque $nIBA = k\phi \Rightarrow \phi = \left(\frac{nAB}{k} \right) I = GI$ or $\phi = GI$... where $[G = \left(\frac{nAB}{k} \right)]$ is the galvanometer constant] $\therefore \phi \propto I$ This provide a linear scale for the galvanometer. (b) Function of a radial magnetic field : Radial magnetic field being normal in all directions is formed to get maximum torque. Function of Soft iron core, which not only makes the field radial but also increases the strength of the magnetic field. (c) One uses a shunt resistance in parallel with the galvanometer, so that most of the current passes through the shunt. In the case of a voltmeter, a resistance of large value is used in series because it must draw a very small current, otherwise the voltage measurement will disturb the original set up by an amount which is very large. OR	$\frac{1}{2} + \frac{1}{2}$ 1 1
--	--	---

(a) **Biot-Savart Law:**

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$$

...where [The terms have the meaning as shown in the figure]

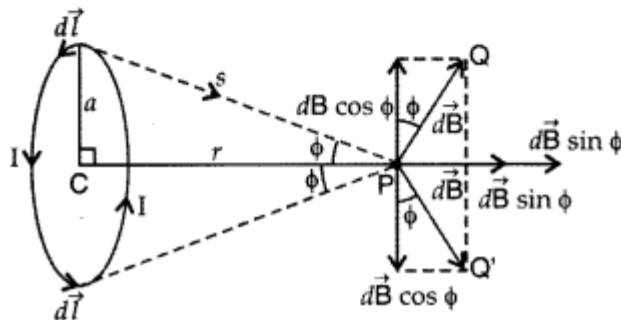


The magnitude $d\vec{B}$ of the magnetic field, due to the element $d\vec{l}$, is given by

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{|d\vec{l} \times \vec{r}|}{r^3} = \frac{\mu_0 I dl}{4\pi r^2}$$

($\because d\vec{l}$ and $\vec{r}$ are $\perp$ to each other)

$$= \frac{\mu_0 I}{4\pi} \frac{dl}{(a^2 + r^2)}$$



The direction of $d\vec{B}$ is shown by vector $P\vec{Q}$.

$d\vec{B}$ can be resolved into two rectangular components,

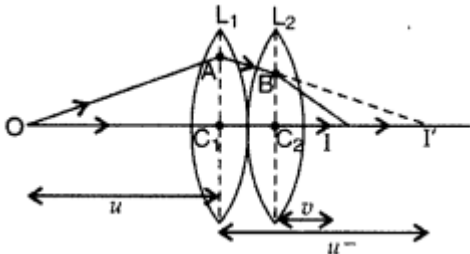
i.e., $d\vec{B} \sin \phi$ along the axis and $d\vec{B} \cos \phi$ perpendicular to the axis.

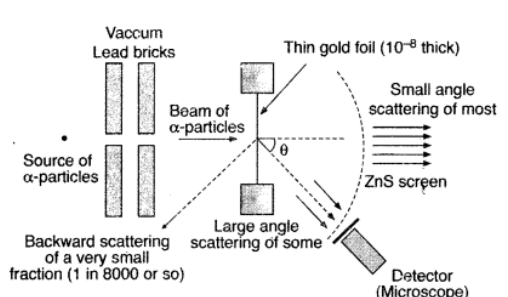
The components perpendicular to the axis of the loop will be equal and opposite to component along the axis of the loop and will cancel out. Their axial components will be in the same direction, i.e., along CP and get added up.

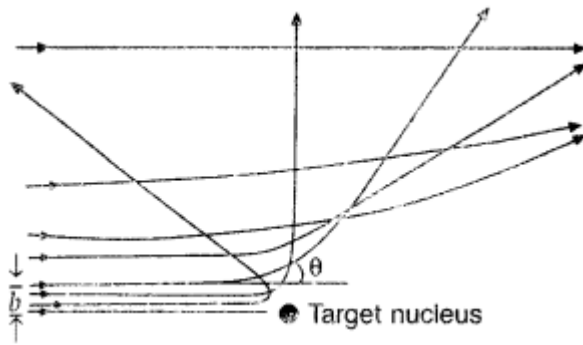
$\therefore$ Total magnetic field at point P in the direction CP is

1

1+1

	Now, to calculate the distance between L_1 and L_2, $u_1 + H_2 = 60 + 15 = 75 \text{ cm}$ Distance between L_2 and $L_3 = v_2 + v_3 = \infty$ or can be any value. OR (a) Power of a lens is defined as the ability to converge a beam of light facing on the lens Since $P = \frac{1}{f}$ Its S.I. unit is dioptre (D) Let C_1, C_2 be the optical centres of two thin convex lenses L_1 and L_2 held co-axially in contact with each other in air.  Suppose f_1 and f_2 are their respective focal lengths. Let a point object O be placed on the common principal axis at a distance $OC_1 = u$. The lens L_1 alone would form its image at I' where $C_1I' = v'$. From the lens formula for L_1, $\frac{1}{v'} - \frac{1}{u} = \frac{1}{f_1} \quad \dots(i)$ I' would serve as a virtual object for lens L_2, which forms a final image I at a distance of $C_2I = v$ As the lenses are thin, therefore, for the lens L_2, $u = C_2I' = C_1I' = v'$ From the lens formula for L_2, $\frac{1}{v} - \frac{1}{v'} = \frac{1}{f_2} \quad \dots(ii)$ Adding equations (i) and (ii), we get	1 1
--	--	-------------------

	$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u}$ Let the two lenses be replaced by a single lens of focal length f which forms image I at distance v, of an object at distance u from the lens For this lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$ $\therefore \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow \frac{1}{F} = \frac{f_1 + f_2}{f_1 f_2} \Rightarrow \boxed{F = \frac{f_1 \times f_2}{f_1 + f_2}}$ (b) (i) For normal adjustment : $m = \frac{f_o}{f_e} = \frac{140}{5} = 28$ (ii) When the final image is formed at the least distance of distinct vision, $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right) = \frac{140}{5} \left(1 + \frac{5}{25} \right)$ $= 28 \left(1 + \frac{1}{5} \right) = 28 \times 1.2 = 33.6$	1 1
33	(a) (i) Geiger-Marsden experiment  (ii) For most of the α-particles, impact parameter is large, hence they suffer very small repulsion due to nucleus and go right through the foil. (iii) Trajectory of α-particles	1 1



It gives an estimate of the size of nucleus, that it relatively very very small as compared to the size of atom.

1

(b) K.E. of the α -particle = Potential energy possessed by beam at distance of closest approach.

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{(2e)(Ze)}{r_0}$$

$$7.7 \times (1.6 \times 10^{-13})$$

$$= \frac{(9 \times 10^9) \times 2 \times (1.6 \times 10^{-19})^2 \times 80}{r_0}$$

1

$$r_0 = \frac{9 \times 10^9 \times 2 \times 2.56 \times 10^{-38} \times 80}{7.7 \times 1.6 \times 10^{-13}} \text{ m}$$

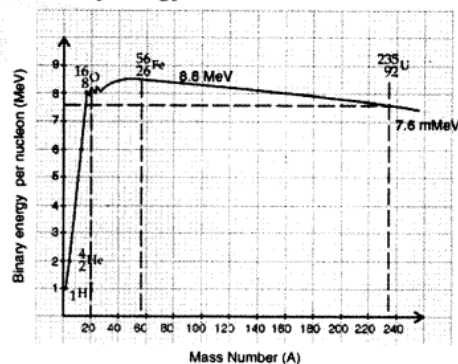
$$= 299 \times 10^{-16} \text{ m}$$

$$= 29.9 \times 10^{-15} \text{ m} \approx 30 \times 10^{-15} \text{ m}$$

1

OR

(a) *Binary Energy Curve*



Conclusions :

(i) The force is attractive and sufficiently strong to produce a binding energy of a few MeV per nucleon.

1

	(ii) For mass number 2 to 20, there are sharply defined peaks corresponding to ${}^2_2\text{He}^4$, ${}^{12}_6\text{C}^{12}$, ${}^{16}_8\text{O}^{16}$ etc. The peak indicates that their nuclei are relatively mass stable than the other nuclei in their neighbourhood.	1/2 +1/2
	(b) (i) When we move from heavy nuclei region to middle region, we find that there will be a gain in overall binding energy and hence release of energy. This indicates that energy can be released when a heavy nucleus breaks into two roughly equal fragments/nuclear fission.	1
	(ii) Similarly, when we move from lighter nuclei to heavier nuclei, we find that there will be gain in overall binding energy and hence release of energy. This indicates that energy can be released when two lighter nuclei fuse together to form heavy nucleus/nuclear fusion.	1/2+1/2
	(c) Nuclear force Any two characteristic properties of nuclear forces	